Yixiao Liu

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Education

University of Pennsylvania Philadelphia, PA
M.S.E. in Mechanical Engineering and Applied Mechanics *Aug. 2025 – Expected Jun. 2027*
- Selected for the SJTU–Penn 3+1+1 Accelerated Master's Program with a one-year tuition waiver.

Shanghai Jiao Tong University Shanghai, China
B.S. in Mechanical Engineering *Aug. 2022 – Jun. 2026*

Research Experience

University of Pennsylvania Feb. 2026 – Present
Research Assistant; Advisor: Jiatao Gu
- Investigating multi-agent collaboration and negotiation in virtual environments.

Institute of Automation, Chinese Academy of Sciences Jun. – Sep. 2026
Research Intern; Advisor: Qiang Liu
- Contributed to idea development, method implementation, experiments and co-writing for **UniForget**, studying concept unlearning across image generation and visual understanding.
- Evaluated BAGEL-7B-MoT and Janus-Pro-7B using joint-forgetting, neighboring-concept retention and capability-preservation metrics; conducted ablation and robustness experiments.

Publications

Is Forgetting Unified? Exploring Concept Unlearning in Unified Multimodal Models

Zhongkai Xue, Junkai Chen, Mingcong Lei, Rumo Zhang, **Yixiao Liu**, Zhimin Sun, Shu Wu, Qiang Liu, Lihua Zhang, Liang Wang.

Submitted to ICLR 2027.

Selected Projects

3D Reconstruction and Neural Rendering Spring 2026
- Implemented calibrated stereo reconstruction with SSD/SAD/ZNCC matching, depth estimation and colored point-cloud export.
- Trained a PyTorch NeRF achieving **24.19 dB PSNR / 0.831 SSIM** across six held-out 100×100 views; compared positional encoding and rendering sample counts.

ROBA Mobile Robot and Embedded Control Fall 2025
- Integrated dual-ESP32 sensing and control, a Python gamepad station, UDP/UART communication, wall following and Vive-based target navigation on a physical competition robot.
- Achieved a **third-place alliance finish** in the Penn MEAM 5100 robotics competition.

UR16e Motion Planning and Control Spring 2025
- Connected collision-checked RRT, smooth trajectory timing and torque-driven PyBullet execution, achieving **0.001179 rad** pooled tracking RMS and **18.349 mm** minimum sampled clearance.

Industry Experience

UAES (United Automotive Electronic Systems) Jun. – Aug. 2025
Software Engineer Intern *Shanghai, China*
- Supported software testing for automotive hardware systems.

Technical Skills

Programming: Python, C/C++, MATLAB, Git
Robotics & Control: PyBullet, RRT, rigid-body dynamics, LQR, MPC, trajectory optimization
Vision & ML: PyTorch, NumPy, SciPy, stereo reconstruction, NeRF, YOLO
Embedded Systems: ESP32, STM32, UART, I^2C, UDP, prototype integration